Aaryan Dhand

+1-587-432-0901 | aaryandhandbusiness@gmail.com | LinkedIn | GitHub | aaryandhand.com

Summary

Software engineering student who moved from data and analytics into building production machine learning. Have shipped a multimodal stress-inference model and a Llama 4 Vision pipeline into live applications, working end to end across data engineering, model training, serving, and integration.

Experience

Vivordo — **Oct 2025 – Present**
AI & Machine Learning Engineer — *Calgary, AB (Remote)*

- Cut **LLM response latency 57%** by re-architecting the **Anthropic**-powered insight layer's inference pipeline, delivering real-time interpretation of biometric signals to users in production.
- Built and shipped a **semi-supervised multimodal stress model** that fuses **HRV**, **Apple Health** biometrics, in-app **chat**, and **mood** check-ins into a calibrated **0–1 stress score**, serving real-time predictions in a live **Flutter + Firebase** app across iOS devices.
- Moved research prototypes into a production analytics product by deploying **TensorFlow/PyTorch** models behind **RESTful APIs** and building the data-preprocessing and model-serving infrastructure they run on.

Hydro One Networks — **May 2025 – May 2026**
Data Analyst Co-op — *Toronto, ON*

- Built the clean data foundation for downstream **ML and predictive analytics** by standing up a centralized **Microsoft SQL Server** warehouse that ingests **SAP** and **OPC** operational data through optimized **SQL** and **Python** pipelines.
- Accelerated material quantification **6x** by leading a **6-person team** to design scalable **Python/VBA ETL workflows** over 300+ historical project records, laying groundwork for forecasting and machine learning.
- Cut manual reporting workload **95%** by engineering end-to-end automation across **SharePoint**, **SAP**, **Excel VBA**, and **Python** for baseline and month-end cost reporting.
- Deployed interactive **Power BI** dashboards wired to live **SAP** data via custom **Python** and **Power Query** pipelines, giving engineering and project leaders real-time cost and inventory visibility.

Projects

SCENR | *Python, PyTorch, CLIP, pgvector, FFmpeg, librosa, PostgreSQL, Supabase*

- Building a **CLIP**-based theme-matching engine for an AI app that turns group-trip media into edited reels, scoring user photos against per-theme fingerprints via **cosine similarity** over a **pgvector** store to keep selected shots on-aesthetic.
- Architecting a deterministic reel pipeline that ranks clips with on-frame **quality/motion models**, beat-syncs cuts via **librosa** onset and tempo detection, and renders themed edits with **FFmpeg** on an async **GPU** queue in under 2 minutes.

FITPIC | *React, Supabase, PostgreSQL, Groq API, Llama 4 Vision, Node.js, Vercel*

- Powered a real-time social analytics platform by building a **computer-vision pipeline** on **Groq's Llama 4 Scout** that extracts structured outfit attributes (color, style, era, formality) from user images with **98% schema-valid output** and **83% tag agreement** on hand-labeled samples, backed by **PostgreSQL** via **Supabase Realtime**.
- Surfaced behavioral and color trends through a personal **analytics dashboard**, secured behind a **serverless** Node.js/Vercel endpoint with **JWT** validation and server-side API-key management.

NBA Betting Odds Analyzer | *Python, Scikit-Learn, XGBoost, Machine Learning*

- Forecast player performance and game outcomes by training and comparing **XGBoost** and **Random Forest** classifiers on predictive features engineered from NBA API game logs merged with live betting-market data.
- Surfaced statistically favorable betting opportunities by backtesting predictions against historical odds and validating models at **0.71 ROC-AUC** with precision-recall analysis.

Education

University of Calgary — **Expected Graduation: May 2027**
Bachelor of Science in Software Engineering; GPA: 3.43 — *Calgary, AB*

- **Awards:** Best Design at TechStart 2025 Final Showcase, Suncor Energy Dependant Scholarship 2022–2026, Jason Lang Scholarship 2023–2026
- **Competitions:** CUSEC 2024, CCPC 2024, NatHacks 2023, Hack the Change 2024
- **Courses:** Data Structures & Algorithms, Object-Oriented Programming, Full Stack Web Development, Computer Architecture, Operating Systems, Networked Systems, Embedded Systems, DevOps, Machine Learning, Software Testing

Technical Skills

Machine Learning & Data: TensorFlow, PyTorch, Scikit-Learn (XGBoost, Random Forest, LSTM, CNNs), ETL Pipelines, REST APIs, SQL Server, Power BI, Power Query
Programming Languages: Python, C, C++, C#, SQL, VBA, JavaScript, HTML/CSS, Assembly
Frameworks & Tools: React, Flask, Django, Unity, Git, Linux, Postman, Jira, Agile, AWS, Google Cloud, Azure DevOps